

Application Build and Release Automation Lab Practice Exam 2

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Problem Scenario

GLA HealthTech is developing a Campus Health Appointment System. Developers are working collaboratively using Linux and Git. Different modules are developed independently before being integrated into the main application. This exam covers managing the project workspace, maintaining version history, developing an independent feature, merging it with the main project, and resolving a conflict created by parallel development.

Question 1 — Linux Project Setup

5 Marks

Create:

- **<FirstName>-GLA-HealthSystem**

Inside the directory create:

- src
- docs
- config
- backup

Create: **health_info.txt**, which must contain:

- Student Name
- Roll Number
- Subject Name
- Project Name
- Semester

Demonstrate appropriate commands to:

- Display current directory.
- Navigate through directories.
- List files.
- Display file contents.
- Display the directory structure.

Answer

```
viyom@Viyom:~$ pwd
/home/viyom
viyom@Viyom:~$ mkdir Viyom-GLA-HealthSystem
viyom@Viyom:~$ cd Viyom-GLA-HealthSystem
viyom@Viyom:~/Viyom-GLA-HealthSystem$ mkdir src docs config backup
viyom@Viyom:~/Viyom-GLA-HealthSystem$ ls
backup  config  docs  src
viyom@Viyom:~/Viyom-GLA-HealthSystem$ touch health_info.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem$ ls
backup  config  docs  health_info.txt  src
viyom@Viyom:~/Viyom-GLA-HealthSystem$ nano health_info.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem$ nano health_info.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem$ cat health_info.txt
Student Name: Viyom Rana
Roll Number: 2415001904
Subject Name: Application Build and Release Automation Lab
Project Name: Campus Health Appointment System
Semester: 5th
viyom@Viyom:~/Viyom-GLA-HealthSystem$ tree
.
├── backup
├── config
├── docs
├── health_info.txt
└── src

5 directories, 1 file
viyom@Viyom:~/Viyom-GLA-HealthSystem$
```

GNU nano 8.7.1 health_info.txt

Student Name: Viyom Rana
Roll Number: 2415001904
Subject Name: Application Build and Release Automation Lab
Project Name: Campus Health Appointment System
Semester: 5th

[Read 5 lines]

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo	M-A Set Mark
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify	^_ Go To Line	M-E Redo	M-G Copy

Question 2 — Git Version Control

6 Marks

- Initialize Git inside the project directory.
- Configure Git identity if required.
- Check Git status.
- Stage the project files.
- Create an initial commit.

Modify health_info.txt by adding:

- Project objective
- Student's program/course
- View the changes made.
- Check Git status.
- Stage the modification.
- Create a second commit.
- Display the commit history.

Answer

```
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/viyom/Viyom-GLA-HealthSystem/.git/
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git config --global user.name "Viyom Rana"
error: invalid key: user.nameViyom Rana
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git config --global user.name "Viyom Rana"
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git config --global user.email "viyomanuragrana@gmail.com"
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git config --global user.name
Viyom Rana
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git config --global user.email
viyomanuragrana@gmail.com
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    health_info.txt

nothing added to commit but untracked files present (use "git add" to track)
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git add health_info.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git status
On branch master

No commits yet
```

```
GNU nano 8.7.1 health_info.txt
Student Name: Viyom Rana
Roll Number: 2415001904
Subject Name: Application Build and Release Automation Lab
Project Name: Campus Health Appointment System
Semester: 5th
Project Objective: To manage campus health appointments efficiently.
Program/Course: B.Tech Computer Science Engineering.

^G Help      ^O Write Out  ^F Where Is   ^K Cut        [ Wrote 7 lines ]
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^T Execute
              ^C Location   ^U Undo       ^M-A Set Mark
              ^_ Go To Line ^E Redo       ^M-G Copy
```

```
Untracked files:
  (use "git add <file>..." to include in what will be committed)
  health_info.txt

nothing added to commit but untracked files present (use "git add" to track)
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git add health_info.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
  new file:   health_info.txt

viyom@Viyom:~/Viyom-GLA-HealthSystem$ git commit -m "Add Health System Information"
[master (root-commit) d297056] Add Health System Information
 1 file changed, 5 insertions(+)
 create mode 100644 health_info.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem$ nano health_info.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem$ get diff
Command 'get' not found, but there are 16 similar ones.
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git diff
diff --git a/health_info.txt b/health_info.txt
index 539e557..40a6227 100644
--- a/health_info.txt
+++ b/health_info.txt
@@ -3,3 +3,5 @@ Roll Number: 2415001904
 Subject Name: Application Build and Release Automation Lab
 Project Name: Campus Health Appointment System
 Semester: 5th
+Project Objective: To manage campus health appointments efficiently.
+Program/Course: B.Tech Computer Science Engineering.
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git status
On branch master

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
```

```
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
  modified:   health_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git add health_info.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git status
On branch master

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
  modified:   health_info.txt

viyom@Viyom:~/Viyom-GLA-HealthSystem$ git commit -m "Update Health System Information"
[master 87fd4d9] Update Health System Information
 1 file changed, 2 insertions(+)
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git log --oneline --all --decorate
87fd4d9 (HEAD -> master) Update Health System Information
d297056 Add Health System Information
viyom@Viyom:~/Viyom-GLA-HealthSystem$ |
```

Question 3 — Independent Feature Development

7 Marks

The team wants to develop an Appointment Module independently.

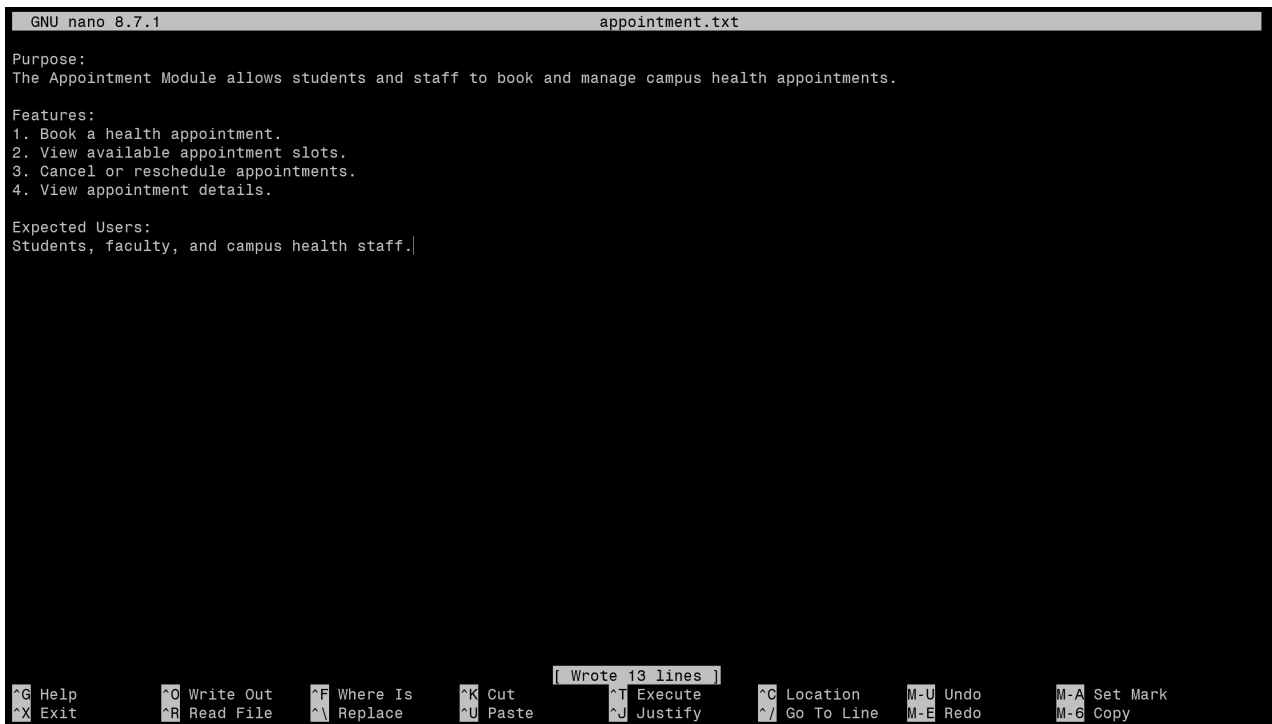
Create a branch named: **appointment-feature**

- Switch to the branch.
- Inside src, create: **appointment.txt**

Add suitable content describing:

- Purpose of the Appointment Module
- At least three features
- Expected users of the module
- Stage and commit the file.
- Display all branches.
- Display the current branch.
- Return to the main branch.
- Demonstrate that the feature work was performed separately.

Answer



```
GNU nano 8.7.1 appointment.txt

Purpose:
The Appointment Module allows students and staff to book and manage campus health appointments.

Features:
1. Book a health appointment.
2. View available appointment slots.
3. Cancel or reschedule appointments.
4. View appointment details.

Expected Users:
Students, faculty, and campus health staff.
```

Wrote 13 lines

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo	M-A Set Mark
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify	^_ Go To Line	M-E Redo	M-G Copy

```

viyom@Viyom:~/Viyom-GLA-HealthSystem$ git branch appointment-feature
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git switch appointment-feature
Switched to branch 'appointment-feature'
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git branch
* appointment-feature
  master
viyom@Viyom:~/Viyom-GLA-HealthSystem$ cd src
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ touch appointment.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ ls
appointment.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ nano appointment.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ cat appointment.txt
Appointment Module

Purpose:
The Appointment Module allows students and staff to book and manage campus health appointments.

Features:
1. Book a health appointment.
2. View available appointment slots.
3. Cancel or reschedule appointments.
4. View appointment details.

Expected Users:
Students, faculty, and campus health staff.
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git status
On branch appointment-feature
Untracked files:
  (use "git add <file>..." to include in what will be committed)
  ./.

nothing added to commit but untracked files present (use "git add" to track)
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git add appointment.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git status
On branch appointment-feature
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)

```

```

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   appointment.txt

viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git commit -m "Add Appointment Module"
[appointment-feature 652fae1] Add Appointment Module
 1 file changed, 13 insertions(+)
 create mode 100644 src/appointment.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git status
On branch appointment-feature
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git branch
* appointment-feature
  master
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git switch master
Switched to branch 'master'
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git status
On branch master
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git show master:src/appointment.txt
fatal: path 'src/appointment.txt' does not exist in 'master'
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git show appointment-feature:/appointment.txt
fatal: path '/appointment.txt' does not exist in 'appointment-feature'
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git show appointment-feature:src/appointment.txt
Appointment Module

Purpose:
The Appointment Module allows students and staff to book and manage campus health appointments.

Features:
1. Book a health appointment.
2. View available appointment slots.
3. Cancel or reschedule appointments.
4. View appointment details.

Expected Users:
Students, faculty, and campus health staff.

```

Question 4 — Merge and Resolve Conflict

8 Marks

Part A – Merge Feature

Merge: **appointment-feature** into the main branch.

- Verify that appointment.txt is now available.

Answer

```
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git log --oneline --all --decorate
652fae1 (HEAD -> master, appointment-feature) Add Appointment Module
87fd4d9 Update Health System Information
d297056 Add Health System Information
```

Part B – Conflict

Create a branch: **system-update**

- Modify a particular line in health_info.txt.
- Commit the change.
- Return to the main branch.
- Modify the same line differently.
- Commit the change.
- Attempt to merge system-update.
- Identify the conflict.
- Open the file and identify the Git conflict markers.
- Resolve the conflict manually.
- Remove the conflict markers.
- Save the final version.
- Stage the resolved file.
- Commit the resolved version.
- Verify the final Git status and commit history.

Answer

```
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git branch system-update
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git switch system-update
Switched to branch 'system-update'
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ nano ../health_info.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ nano ../health_info.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ nano ../health_info.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git add ../health_info.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git status
On branch system-update
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   ../health_info.txt

viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git commit -m "Update health system objective on system-update branch"
[system-update c2e0c01] Update health system objective on system-update branch
 1 file changed, 1 insertion(+), 1 deletion(-)
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git switch master
Switched to branch 'master'
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ nano ../health_info.txt
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ |
```

```
GNU nano 8.7.1 ../health_info.txt
Student Name: Viyom Rana
Roll Number: 2415001904
Subject Name: Application Build and Release Automation Lab
Project Name: Campus Health Appointment System
Semester: 5th
Project Objective: To provide an efficient campus health appointment system with online booking and appointment management.
Program/Course: B.Tech Computer Science Engineering.

[ Read 7 lines ]
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line M-E Redo     M-6 Copy
```

```
GNU nano 8.7.1 ../health_info.txt
Student Name: Viyom Rana
Roll Number: 2415001904
Subject Name: Application Build and Release Automation Lab
Project Name: Campus Health Appointment System
Semester: 5th
Project Objective: To simplify campus healthcare access by allowing users to schedule and manage health appointments.
Program/Course: B.Tech Computer Science Engineering.

[ Wrote 7 lines ]
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line M-E Redo     M-6 Copy
```

```
GNU nano 8.7.1 ../health_info.txt
Student Name: Viyom Rana
Roll Number: 2415001904
Subject Name: Application Build and Release Automation Lab
Project Name: Campus Health Appointment System
Semester: 5th
<<<<<<< HEAD
Project Objective: To simplify campus healthcare access by allowing users to schedule and manage health appointments.
=====
Project Objective: To provide an efficient campus health appointment system with online booking and appointment management.
>>>>>>> system-update
Program/Course: B.Tech Computer Science Engineering.

[ Read 11 lines ]
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line M-E Redo     M-6 Copy
```

```
viiom@viiom:~/Viiom-GLA-HealthSystem/src$ git switch master
Switched to branch 'master'
viiom@viiom:~/Viiom-GLA-HealthSystem/src$ nano ../health_info.txt
viiom@viiom:~/Viiom-GLA-HealthSystem/src$ git add ../health_info.txt
viiom@viiom:~/Viiom-GLA-HealthSystem/src$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   ../health_info.txt

viiom@viiom:~/Viiom-GLA-HealthSystem/src$ git commit -m"Update health system objective on master branch"
[master e859b3f] Update health system objective on master branch
 1 file changed, 1 insertion(+), 1 deletion(-)
viiom@viiom:~/Viiom-GLA-HealthSystem/src$ git merge system-update
Auto-merging health_info.txt
CONFLICT (content): Merge conflict in health_info.txt
Automatic merge failed; fix conflicts and then commit the result.
viiom@viiom:~/Viiom-GLA-HealthSystem/src$ git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   ../health_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
viiom@viiom:~/Viiom-GLA-HealthSystem/src$ nano ../health_info.txt
viiom@viiom:~/Viiom-GLA-HealthSystem/src$ git add ../health_info.txt
viiom@viiom:~/Viiom-GLA-HealthSystem/src$ git status
On branch master
All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)
```

```
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git commit -m"Resolve health system merge conflict"
[master be25a89] Resolve health system merge conflict
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git status
On branch master
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git log --oneline --all --decorate
be25a89 (HEAD -> master) Resolve health system merge conflict
e859b3f Update health system objective on master branch
c2e0c01 (system-update) Update health system objective on system-update branch
652fae1 (appointment-feature) Add Appointment Module
87fd4d9 Update Health System Information
d297056 Add Health System Information
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ |
```

Question 5 — Repository Demonstration

4 Marks

Demonstrate:

- Project directory structure.
- Git repository.
- Commit history.
- Feature branch.
- Successful merge.
- Merge conflict and resolution.

Answer

```
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ git log --oneline --all --decorate
be25a89 (HEAD -> master) Resolve health system merge conflict
e859b3f Update health system objective on master branch
c2e0c01 (system-update) Update health system objective on system-update branch
652fae1 (appointment-feature) Add Appointment Module
87fd4d9 Update Health System Information
d297056 Add Health System Information
viyom@Viyom:~/Viyom-GLA-HealthSystem/src$ cd ..
viyom@Viyom:~/Viyom-GLA-HealthSystem$ tree
.
├── backup
├── config
├── docs
├── health_info.txt
├── src
│   └── appointment.txt
└──
5 directories, 2 files
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git status
On branch master
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git branch
* master
  system-update
viyom@Viyom:~/Viyom-GLA-HealthSystem$ git log --oneline --all --decorate
be25a89 (HEAD -> master) Resolve health system merge conflict
e859b3f Update health system objective on master branch
c2e0c01 (system-update) Update health system objective on system-update branch
652fae1 (appointment-feature) Add Appointment Module
87fd4d9 Update Health System Information
d297056 Add Health System Information
viyom@Viyom:~/Viyom-GLA-HealthSystem$ |
```